

Computer Modelling & Simulation

Section # 6

Problem 1

Derive the CDF of $\text{Exp}(\mu)$ and $U(a,b)$, also derive equation for the Random Observation (R.O) of above distribution.

Answer

- Uniform distribution $U(a,b)$:

CDF:-

$$\text{pdf : } f(x) = \frac{1}{b-a} \quad a \leq x \leq b$$

$$\text{Then CDF : } F(x) = \int_a^x \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^x dx = \frac{1}{b-a} \cdot x \Big|_a^x$$

$$F(x) - F(a) = \frac{x}{b-a} - \frac{a}{b-a}$$

$$F(x) = \frac{x-a}{b-a} \quad a \leq x \leq b$$

 $\longrightarrow$ "CDF"

R.O equation:-

$$R = F(x)$$

$$R = \frac{x-a}{b-a}$$

$$x - a = R(b - a)$$

$$x = R(b - a) + a \quad 0 \leq R \leq 1$$

 $\longrightarrow$ "R.O"

- Exponential distribution $\text{Exp}(\mu)$:

CDF:-

$$\text{pdf : } f(x) = \mu e^{-\mu x} \quad 0 \leq x \leq \infty$$

Then CDF : $F(X) = \int_0^x \mu e^{-\mu x} dx = \mu \int_0^x e^{-\mu x} dx = \mu \cdot \frac{e^{-\mu x}}{-\mu} \Big|_0^x = -e^{-\mu x} \Big|_0^x$

$$F(X) = F(x) - F(0) = -e^{-\mu x} - (-1) = -e^{-\mu x} + 1 \quad 0 \leq x \leq \infty$$

$$F(x) = 1 - e^{-\mu x} \quad \longrightarrow \quad \text{"CDF"}$$

$$\begin{aligned} R &= 1 - e^{-\mu x} \\ e^{-\mu x} &= 1 - R \\ \ln e^{-\mu x} &= \ln(1 - R) \\ -\mu x &= \ln(1 - R) \quad \text{-----} \end{aligned}$$

$$x = \frac{\ln(1-R)}{-\mu}, \quad 0 \leq R \leq 1 \quad \longrightarrow \quad \text{"R.O"}$$

Problem 2

- Generate 4 R.O from the following distribution: the random variables has $p(x=0)=0.4$, given $x \neq 0$ it has a uniform distribution between -25 and 20. Use the following Random Numbers 1,5,3,7,4,2,1,1
- Generate 4 R.N observation from a variable follows a $U(40,50)$ with probability 0.5 else it has a uniform distribution between -25 and 25 . use the following R.N:125,505,553,997,117,009,888,444

Answer

- R.N $0 \rightarrow 9$
 $P(x=0) = 0.4 \quad 0 \rightarrow 3$
 $P(x \neq 0) = 0.6 \quad 4 \rightarrow 9$
 $U(-25,20)$
 $X = R(b-a) + a \quad 0 \leq R \leq 1$
 $X = R(20+25) - 25$
 $X = 45R - 25, \quad 0 \leq R \leq 1$
 $\rightarrow$ R.O:
 1) RN=1, $X_1 = 0$
 2) RN=5, $X_2 \neq 0$
 $RN=3, X_2 = 45.3/10 - 25 = -23/2 = -11.5$
 3) RN=7, $X_3 \neq 0$
 $RN=4, X_3 = 45.4/10 - 25 = -7$
 4) RN=2, $X_4 = 0$
- $U(40,50) \rightarrow 0.5 \quad 000 \rightarrow 499, U(-25,25) \rightarrow 0.5 \quad 500 \rightarrow 999$
 When $U(40,50)$

$$X=R(b-a) + a \quad a \leq x \leq b$$

$$X=R(50-40) + 40$$

$$X=10R+40 \quad 0 \leq R \leq 1, 40 \leq x \leq 50$$

When $U(-25,25)$

$$X=R(b-a) + a \quad a \leq x \leq b$$

$$X=R(25 + 25) -25$$

$$X=50R - 25 \quad 0 \leq R \leq 1, -25 \leq x \leq 25$$

To get 4 random observations:

$$1) R=125 \quad 000 \rightarrow 499$$

$$R= 505/1000, R \text{ must be in that range } 0 \leq R \leq 1$$

$$X=10R+40$$

$$X_1= (10 *505/1000) + 40= 45.05$$

x belongs to in that range $40 \leq x \leq 50$

$$2) R=553 \quad 500 \rightarrow 999$$

$$R= 997/1000, R \text{ must be in that range } 0 \leq R \leq 1$$

$$X=50 R - 25$$

$$X_2=(50 *997/1000) - 25= 24.85$$

x belongs to that range $-25 \leq x \leq 25$

$$3) R=117 \quad 000 \rightarrow 499$$

$$R= 009/1000, R \text{ must be in that range } 0 \leq R \leq 1$$

$$X=10R+40$$

$$X_3=10* +40= 40.09$$

x belongs to that range $40 \leq x \leq 50$

$$4) R= 888$$

$$R= 444/1000 \quad R \text{ must be in that range } 0 \leq R \leq 1$$

$$X=50R - 25$$

$$X_4= (50 * 444/1000) - 25 = -2.8$$

x belongs to that range $-25 \leq x \leq 25$